

Assingment-6

Task Description-1 (Classes – Data Validation)

Prompt: generate a Student class with attributes: name, roll_no, and marks. Add a method is_pass() that returns whether the student has passed (marks $\geq$ 40).

Code:

```
class Student:

    def __init__(self, name, roll_no, marks):

        self.name = name

        self.roll_no = roll_no

        self.marks = marks

    def is_pass(self):

        return self.marks >= 40

#example usage

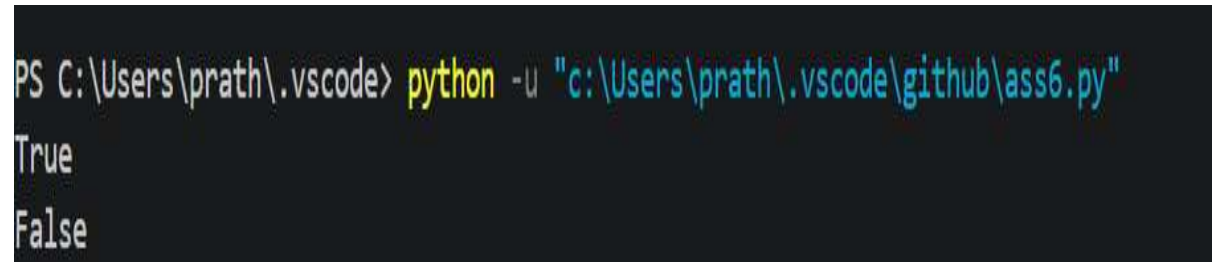
student1 = Student("Alice", 1, 75)

student2 = Student("Bob", 2, 35)

print(student1.is_pass()) # Output: True

print(student2.is_pass()) # Output: False
```

output:



```
PS C:\Users\prath\.vscode> python -u "c:\Users\prath\.vscode\github\ass6.py"
True
False
```

Task Description-2 (Loops – Pattern Generation)

Prompt: generate a function that prints a right-angled triangle star pattern using a for loop. Then regenerate the same pattern using a while loop.

Code:

```
def print_triangle_for_loop(rows):
```

```
    for i in range(1, rows + 1):
```

```
        print("*" * i)
```

#Here is a function that prints a right-angled triangle star pattern using a while loop:

```
def print_triangle_while_loop(rows):
```

```
    i = 1
```

```
    while i <= rows:
```

```
        print("*" * i)
```

```
        i += 1
```

#example usage

```
print("Triangle using for loop:")
```

```
print_triangle_for_loop(5)
```

```
print("Triangle using while loop:")
```

```
print_triangle_while_loop(5)
```

Output:

```
PS C:\Users\prath\.vscode> python -u "c:\Users\prath\.vscode\github\ass6.py"
Triangle using for loop:
*
**
***
****
*****
Triangle using while loop:
*
**
***
****
*****
```

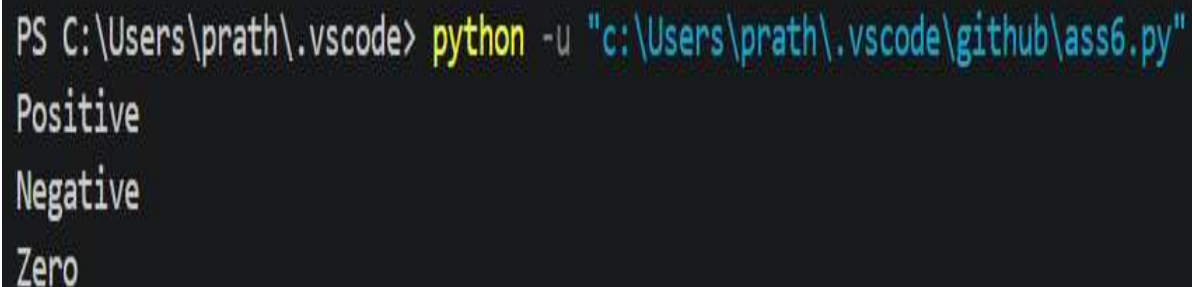
Task Description-3 (Conditional Statements – Number Analysis)

Prompt: write a function that checks whether a given number is positive, negative, or zero using if-elif-else. Test the function with multiple inputs.

Code:

```
def check_number_sign(number):  
    if number > 0:  
        return "Positive"  
    elif number < 0:  
        return "Negative"  
    else:  
        return "Zero"  
  
#example usage  
print(check_number_sign(5)) # Output: Positive  
print(check_number_sign(-3)) # Output: Negative  
print(check_number_sign(0)) # Output: Zero
```

output:

A screenshot of a Windows command prompt window. The prompt is 'PS C:\Users\prath\.vscode>'. The command entered is 'python -u "c:\Users\prath\.vscode\github\ass6.py"'. The output of the script is displayed on three separate lines: 'Positive', 'Negative', and 'Zero'.

```
PS C:\Users\prath\.vscode> python -u "c:\Users\prath\.vscode\github\ass6.py"  
Positive  
Negative  
Zero
```

Task Description-4 (Nested Conditionals)

Prompt: Generate a function `check_discount(age, is_member)` that determines discount eligibility:

- Age $\geq 60 \rightarrow$ Senior discount
- Member $\rightarrow$ Additional discount

Use nested if statements.

Code:

```
def check_discount(age, is_member):  
    if age >= 60:  
        discount = "Senior discount"  
        if is_member:  
            discount += " + Additional member discount"  
    else:  
        discount = "No senior discount"  
        if is_member:  
            discount += " + Additional member discount"  
    return discount  
  
#example usage  
  
print(check_discount(65, True)) # Output: Senior discount + Additional member  
discount  
  
print(check_discount(50, True)) # Output: No senior discount + Additional member  
discount  
  
print(check_discount(65, False)) # Output: Senior discount
```

output:

```
PS C:\Users\prath\.vscode> python -u "c:\Users\prath\.vscode\github\ass6.py"  
Senior discount + Additional member discount  
No senior discount + Additional member discount  
Senior discount
```

Task Description-5 (Class – Mathematical Opera)

Prompt: create a Circle class with methods to calculate area () and circumference () given the radius.

Code:

```
import math

class Circle:

    def __init__(self, radius):

        self.radius = radius

    def area(self):

        return math.pi * self.radius ** 2

    def circumference(self):

        return 2 * math.pi * self.radius

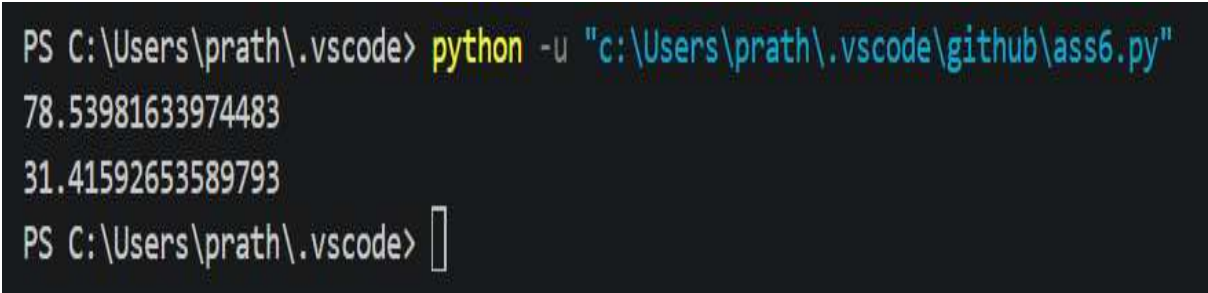
#example usage

circle = Circle(5)

print(circle.area())    # Output: 78.53981633974483

print(circle.circumference()) # Output: 31.41592653589793
```

Output:



```
PS C:\Users\prath\.vscode> python -u "c:\Users\prath\.vscode\github\ass6.py"
78.53981633974483
31.41592653589793
PS C:\Users\prath\.vscode> █
```